

UEC812: WIRELESS SENSOR NETWORKS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To provide a succinct introduction to the field of wireless sensor networks by introducing the fundamentals of network architectures, protocols and deployment methods. To familiarise with various networks platforms and tools for wireless sensor networks				
Syllabus Introduction and Overview of Wireless Sensor Networks: Background of Sensor Network Technology, Application of Sensor Networks, Challenges for Wireless Sensor Networks, Enabling Technologies for Wireless Sensor Networks. Sensor Node Hardware and Network Architecture: Single-node Architecture: Hardware Components, Operating Systems and Execution Environments, Network Architecture: Sensor Network Scenarios, Optimization Goals and Figures of Merit, Gateway Concepts. Network Protocols: MAC Protocols: Requirement and design constraints for MAC Protocols, Important classes of MAC Protocols, MAC Protocols for Wireless Sensor Networks, Routing Protocols: Classification of Routing Protocols, Energy-Efficient Routing, Geographic Routing. Deployment and Configuration: Localization and Positioning, Single-hop Localization, Positioning in Multi-hop environments, Coverage and Connectivity, Naming and Addressing in Sensor Networks, Assignment of MAC addresses. Sensor Network Platforms and Tools: Sensor Node Hardware – Berkeley Motes, Programming Challenges, Node-level Software Platforms, Node-level Simulators. Familiarization with standards: <i>IEEE 1451.0 and IEEE 1451.5-802.11.</i> Micro Project: WSN based monitoring of Temperature				
Laboratory Work: Introduction to Python, Development of a neural network, Apply gradient descent algorithm, Explore LeNet, Use pertained models for recognition of MINST/CIFAR-10.				
Course Learning Objectives (CLO) The students will be able to: 1. Define the fundamental concepts of Wireless Sensor Networks (WSNs) and list their applications in various domains. 2. Explain the various architectures of Wireless Sensor Networks, its related hardware and protocols 3. Familiarise with deployment and configuration methods. 4. Acquainted to Node-level Software Platforms.				
Text Books 1. Holger Karland Andreas Willig, <i>Protocols and Architectures for Wireless Sensor Networks</i>, JohnWiley,2005. 2. Feng Zhao & Leonidas J. Guibas, <i>Wireless Sensor Networks - An Information Processing Approach</i>, Elsevier,2007.				

Reference Books

1. *Kazem Sohraby, Daniel Minoli, and Taieb Znati, Wireless Sensor Networks-Technology, Protocols and Applications, John Wiley, (2007).*
2. *Raghavendra, Cauligi S, Sivalingam, Krishna M., Zanti Taieb, Wireless Sensor Network, Springer 1st Ed, (2004) (ISBN: 978-4020-7883-5).*
3. *Anna Hac, Wireless Sensor Network Designs, John Wiley, (2003).*